
Multisections of 4-Manifolds

Computing Invariants of 4-Manifolds using Curves on Surfaces

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The Basics

Heegaard Splittings

Trisections and Multisections

Computing Intersection Forms using Multisections

Results on $(2, n)$ -Sections from $\mathbb{Z}HS^3$'s

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- ▶ We study manifolds up to *diffeomorphism*, or smooth manipulations that don't rip or tear our space and which can be undone without ripping or tearing

Some Examples

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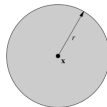
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- ▶ $B^n = \{v \in \mathbb{R}^n : \|v\| \leq 1\}$ and their boundaries
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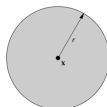
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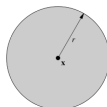


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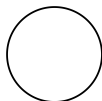
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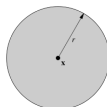
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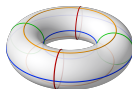
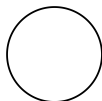
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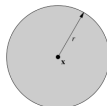
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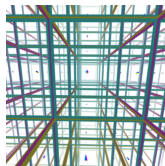
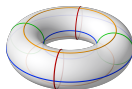
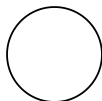
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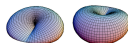
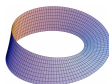
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- ▶ Spaces which are not locally trivial or have singularities

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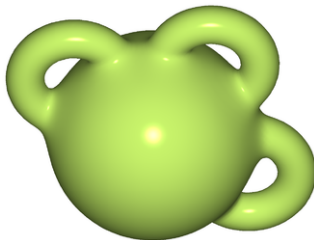
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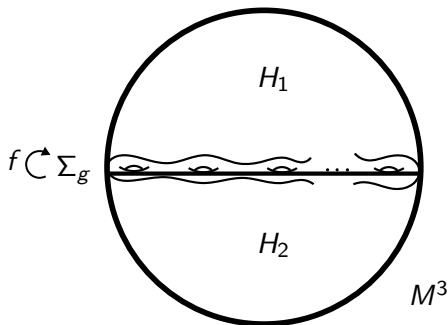
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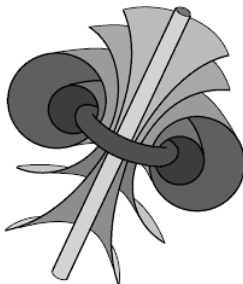
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- ▶ This gives our genus 1 Heegaard splitting of S^3



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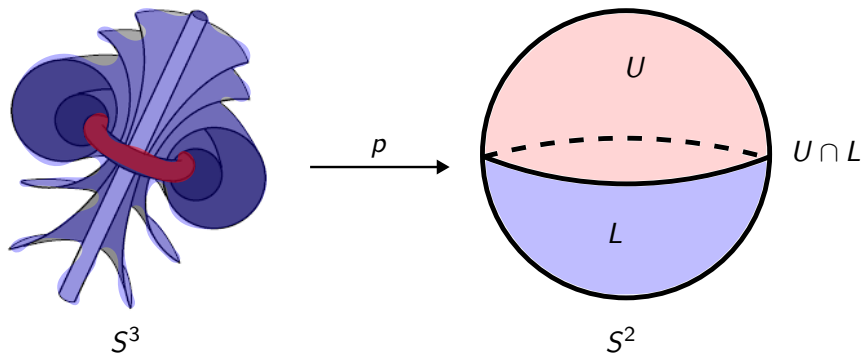


Figure: Hopf fibration with preimages of U and L

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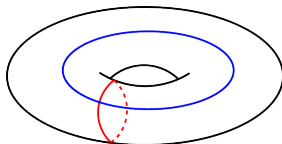
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- ▶ The diagram tells us to take one genus 1 handlebody and attach another such that the red curve is sent to the blue curve



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If (Σ_g, H_1, H_2) is a Heegaard splitting, a compatible *genus g Heegaard diagram* consists of sets of attaching circles $\alpha = \{\alpha_1, \dots, \alpha_g\}$ and $\beta = \{\beta_1, \dots, \beta_g\}$ on Σ_g for H_1, H_2 , resp.

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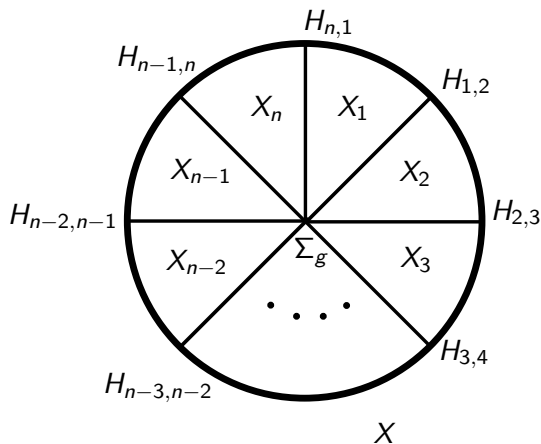
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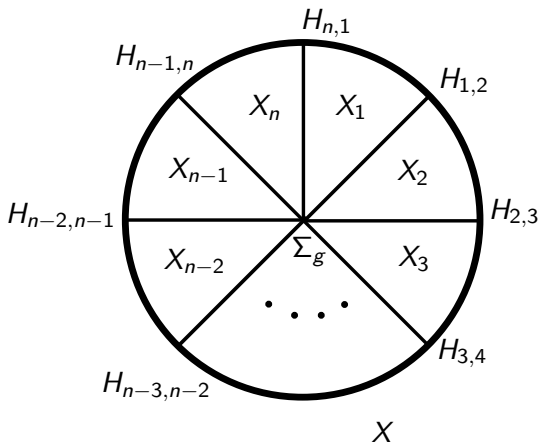
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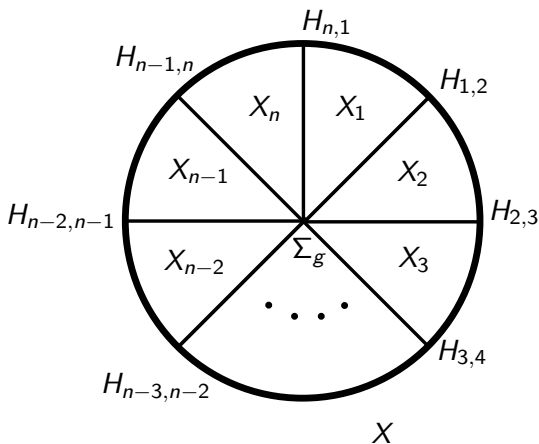
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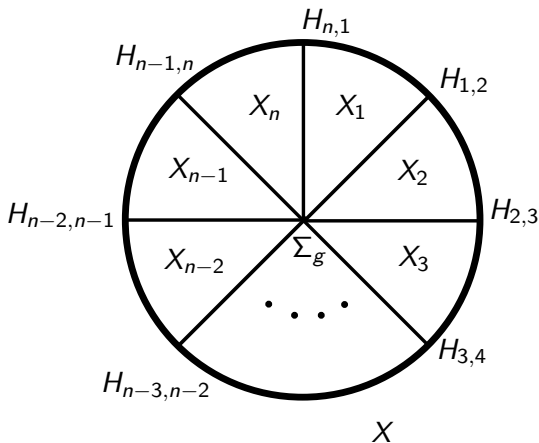
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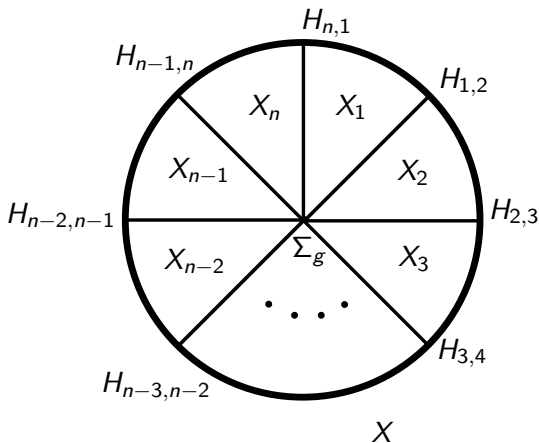
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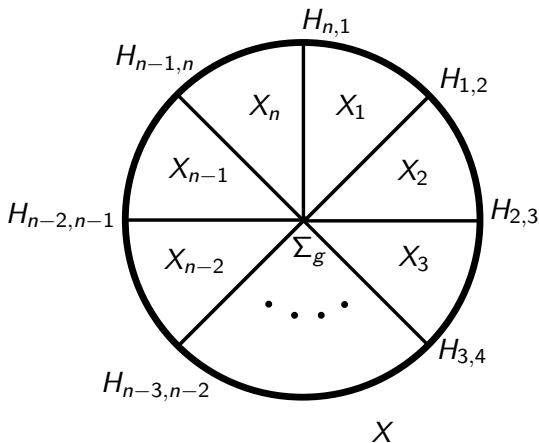
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- ▶ Else $X_i \cap X_j = \Sigma_g$
- ▶ $H_{i-1,i} \cup_{\Sigma_g} H_{i,i+1}$ is a Heegaard splitting for $\partial X_i \cong \#^{k_i} S^1 \times S^2$

Using Handles to Study 4-Manifolds

Theorem 2

Every compact oriented 4-manifold admits a trisection decomposition

Trisections and Multisections

Example: $\mathbb{CP}^2$

¹ This is called the *momentum map* with respect to the standard Kähler form for the torus action given by

$$[z_1 : z_2 : 1] \mapsto [e^{i\theta} z_1 : e^{i\varphi} z_2 : 1]$$

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Define $\mu : \mathbb{CP}^2 \rightarrow \mathbb{R}^2$ as¹

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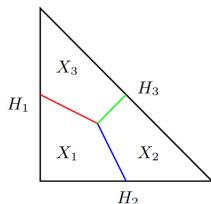
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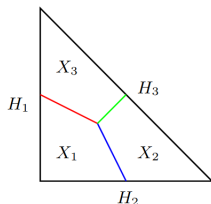
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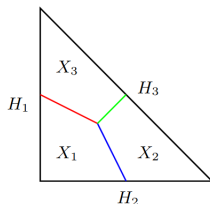


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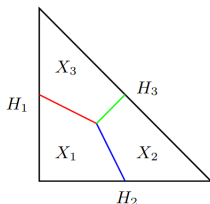


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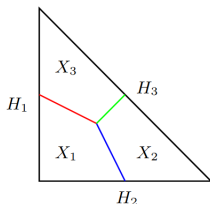


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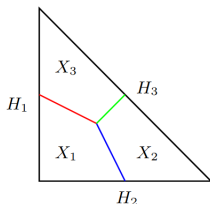


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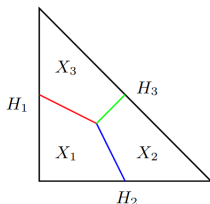


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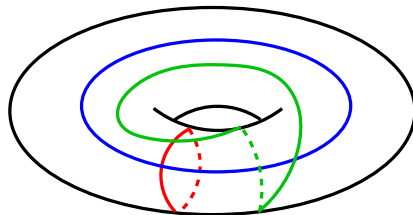
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 - So this is a genus 1 trisection of $\mathbb{CP}^2$

Trisection Diagrams

We can express multisections just as we did Heegaard diagrams, with curves on surfaces

Example

Trisection diagram for $\mathbb{CP}^2$



Multisection Diagrams

Definition

A (g, k, n) -section diagram is $(\Sigma_g, \gamma_1, \dots, \gamma_n)$ such that $(\Sigma_g, \gamma_i, \gamma_{i+1})$ are genus g Heegaard diagrams for $\#^k(S^1 \times S^2)$

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- For a fixed 4-manifold X , we have that k is determined by X and g , and so we will freely refer to (g, n) -sections of X

The Basics

Heegaard Splittings

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Results on $(2, n)$ -Sections from $\mathbb{Z}HS^3$'s

Intersection Forms

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 - This count is finite since $\dim \Sigma_1 + \dim \Sigma_2 = \dim X$ and the intersection is transverse

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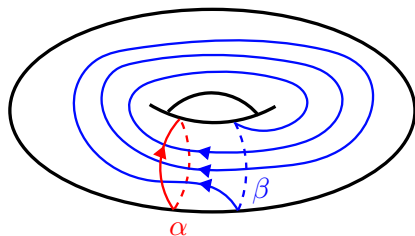
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$$\langle \alpha, \beta \rangle_{T^2} = +3$$

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$$\partial_2((x_i)_{1 \leq i \leq n}) = ((x_i - x_{i+1})_{1 \leq i \leq n}), \quad \partial_1((x_i)_{1 \leq i \leq n}) = \sum_{i=1}^n x_i$$

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Moreover, Q_X can be represented as a matrix Q with entries depending only on the evaluation of $\langle -, - \rangle_{\Sigma_g}$ on a basis γ_i, γ_j for complementary Lagrangians

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$$E_8 = \begin{pmatrix} 2 & 1 & 1 & 0 & 1 & 0 & 0 & 0 \\ 1 & 2 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 2 & 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 2 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 2 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 2 \end{pmatrix}$$